# Himanshu Mahajan

Fullstack Engineer (Python/React)

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Software Engineer (~2 years exp.) specialized in building scalable ETL/ELT pipelines, concurrent backend services, and automated data processing systems. Expert in Python, Unix/Bash scripting, and relational database engineering (PostgreSQL/SQL), with proven experience orchestrating cloud-hosted integrations (GCP/AWS).

## SKILLS

- **Tech Stack & Concepts:** Python, JS (ES6+), SQL, Unix/Bash Scripting, OOP, SDLC, RESTful APIs, ETL/ELT Pipelines
- **Cloud & DevOps:** Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Git, CI/CD, Celery, Sentry
- **Databases & Caching:** PostgreSQL, MySQL, SQLite, MongoDB, Firebase, Redis, pgvector

## WORK EXPERIENCE

**Software Engineer (Full-Stack & Data),** *Patils Classes* 04/2026 – Present
Nagpur

- Engineered a pgvector & PostgreSQL Question Bank; built a **RAG-based semantic-search engine** using Cosine Similarity to recommend practice questions based on topic gaps.
- Developed an automated document OCR pipeline via Docling & PaddleOCR; orchestrated local VLMs (Ollama) and used **Pydantic** to parse and validate mathematical outputs into structured PostgreSQL.
- Launched a Flask/Firebase CBT portal (patilsclasses.com/test ) with canvas question mapping and an automated window-switching anti-cheat.
- Built a serverless corporate website (patilsclassesai.com ) and portal using **Next.js**, React, Cloudflare Workers, and D1 DB to manage integrations.

**Software Engineer,** *Delphi Analytics* 04/2025 – 12/2025
Nagpur

- Designed Python client SDKs using Object-Oriented Programming (OOP) to standardize RESTful microservices, reducing integration overhead by 40%.
- Owned DevEx and code reviews by authoring robust Swagger/OpenAPI specifications, accelerating developer onboarding.
- Leveraged AsyncIO, Flask, and Redis to scale high-throughput RESTful APIs, optimizing real-time data streaming over WebSockets on Linux.
- Designed relational SQLAlchemy ORM models in PostgreSQL, optimizing query performance and configuring automated database backup and disaster recovery (DR) protocols.

**Software Engineering Intern,** *Delphi Analytics* 08/2024 – 03/2025
Nagpur

- Migrated legacy BI to Apache Superset, designing robust **ETL/ELT pipelines** to process complex operational datasets.
- Collaborated with UX designers and data engineers to build interactive dashboards using React and Superset, reducing reporting latency.

## EDUCATION

**B.Tech in Metallurgical and Materials Engineering,** 2024
*Visvesvaraya National Institute of Technology* Nagpur, India
Relevant Coursework: DSA, SQL Database Design, Applied Data Analytics

## PROJECTS

**Gym SaaS Multi-Tenant Platform (Django, PostgreSQL, Celery, Redis)**
Built a multi-tenant platform featuring schema-level database isolation, secure Stripe API payment integrations, and background billing workflows orchestrated via Celery (distributed task queue).

**CricAudit Event-Sourcing Ledger (Python, SQLite, Streamlit, Pydantic)**
Developed an analytical engine validating ledger invariants and reconstructing match states from ball-by-ball delivery logs via stream-replay event sourcing.